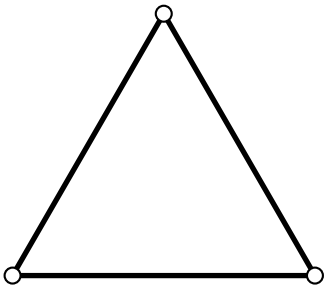
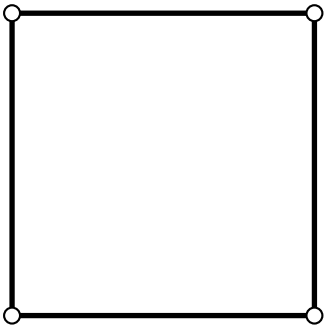
Keep the frames flat. Bars stay straight and keep their length; joints may turn. Sliding or turning the whole frame does not change its shape.

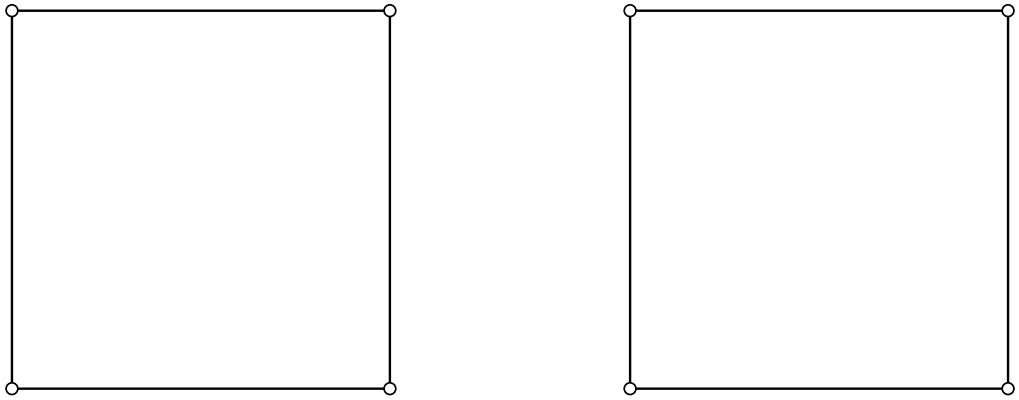
Problem 1: Can you change the shape of the triangle and the square? Draw the shapes you can make.





Problem 2: Make the square hold its shape with the fewest diagonal bars. Draw where you put the bars.

A diagonal bar joins opposite corner pins. It is called a brace.

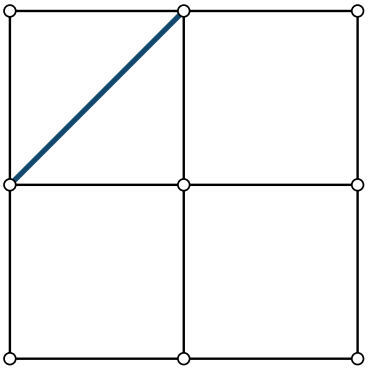


Problem 3: Remove the diagonal and make shapes with the four side bars. Which shapes can you fit that same diagonal into?

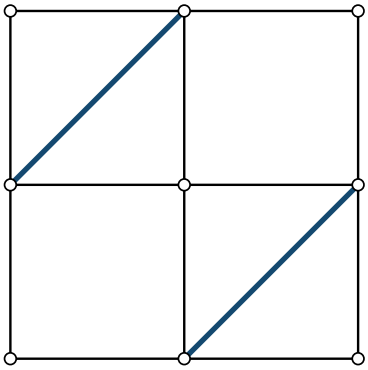
Keep the frames flat. Moving or turning the whole frame does not change its shape. In a grid, use at most one diagonal brace in each cell.

Problem 4: Build these designs on your 2-by-2 frame. Circle each design that holds its shape. For each frame that can change shape, draw a changed shape.

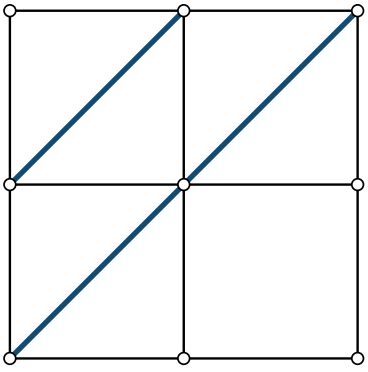
A



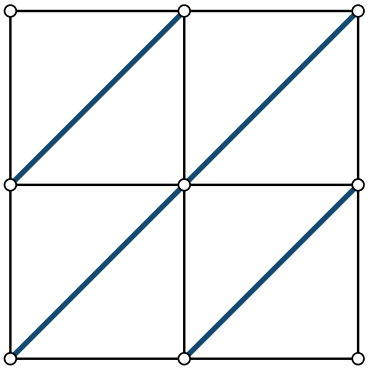
B

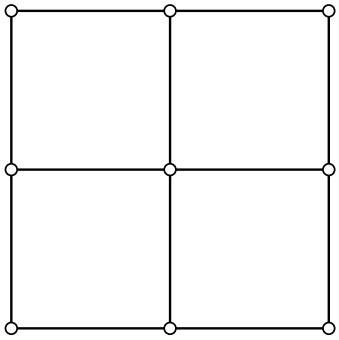
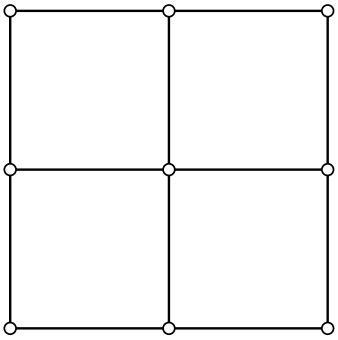
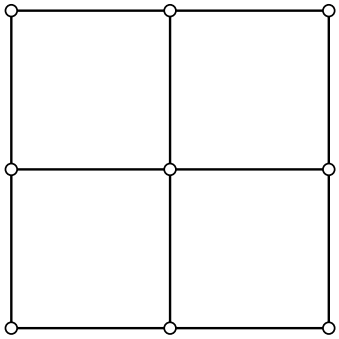
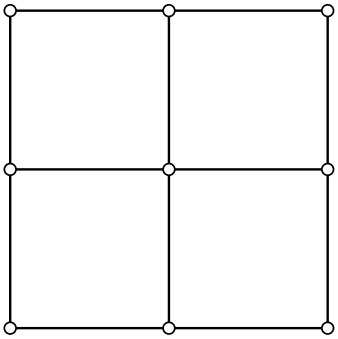


C



D



Problem 5: Find the fewest braces that make your 2-by-2 frame hold its shape. Find several designs using that many braces.

Fewest braces: _____